Tarun Mittal

Senior Software Engineer

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Summary

- Senior Software Engineer with **5 years of experience** building high-concurrency distributed backends and production **AI/agentic systems** across retail-tech, fintech, and marketplace domains.
- Polyglot engineer – **Python, Go, Rust** – deep in event-driven systems, streaming (**Kafka, Flink**), OLAP (**ClickHouse, BigQuery**), and agent orchestration (**LangGraph, MCP**).

Skills

- **Languages**: Python, Go (Golang), Rust, SQL, JavaScript/TypeScript.
- **Backend & APIs**: FastAPI, Gin, Axum, Django, SQLAlchemy, Pydantic, REST, gRPC, WebSockets, SSE, Celery.
- **AI & Agents**: LangGraph, LangChain, MCP, RAG, text-to-SQL guardrails, semantic caching, pgvector, OpenAI / Gemini / Claude APIs.
- **Streaming & Data**: Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, ClickHouse, BigQuery, DuckDB.
- **Databases, Cloud & Infra**: PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, GCP, AWS, Docker, Kubernetes, CI/CD, ELK, Grafana, New Relic.
- **Core Engineering**: Data Structures & Algorithms, System Design (HLD/LLD), Concurrency, DBMS, OS, Networking, Unit Testing.

Professional Experience

Senior Software Engineer | IMPACT ANALYTICS, Bangalore | June 2026 – Present

Agentic Assort Planner — Multi-Agent Retail Merchandise-Planning Platform

- Architected a router→orchestrator→specialist **multi-agent system (5 domain agents)** decomposing natural-language planning questions into a validated execution DAG, **raising decomposition quality 0.54 → 0.99** (offline eval) via a structural DAG validator with one LLM self-repair pass.
- Designed the polyglot service architecture — **Go (Gin) for all non-agentic services, Rust (Axum) for performance-critical APIs**, Python (LangGraph, MCP) for agents.
- Led adoption of **ClickHouse as the end-to-end transactional + analytical planning store** by designing an insert-only, versioned write model (ReplacingMergeTree, argMax reads, atomic partition swaps) that **holds the async-mutation queue at zero at 10× scale** while natively powering overrides, version-diff, and undo/redo.
- Cut recurring **BigQuery scan cost up to 280× (2.9 TiB → 3.3 GiB/query)** via clustered-copy routing, a store×category×week rollup, and a query-path cost guard — after catching one SELECT * query billing 38.8 TiB ($327) for 2 rows.
- **Tech Used**: Python, FastAPI, LangGraph, MCP, Go, Gin, Rust, Axum, ClickHouse, BigQuery, pgvector, Redis, Kafka, GCP.

Agentic Store-Clustering Copilot (flagship module)

- Designed an agentic clustering copilot — deterministic tools on a dedicated ClickHouse read plane, LLM orchestration, three human confirm-gates — targeting **under 1 h to a finalized plan (from days), 20+ configs explored (from 1), and run failures below 2% (from 8.5%).**
- Cut agent data-probe latency from **1–20 s (shared BigQuery slots) to p95 under 500 ms** via a dedicated ClickHouse read plane with nightly precompute (80%+ warm sessions).
- Eliminated the input-error class behind **80%+ of run failures** via deterministic grounding (catalog backtracking, season resolution, confirm gates); made shipped clusterings **100% reproducible (from 0%)** via content-addressed config documents and durable pins.
- **Tech Used**: Python, LangGraph, MCP, ClickHouse, BigQuery, scikit-learn, FastAPI, SSE, GCP.

Software Engineer | UBER (via EPAM Systems), Bangalore | July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform — Uber Finance

- Cut Uber's quarterly financial-risk-scoping cycle time by **70%** by replacing a manual Google-Sheets workflow with a FastAPI + MySQL platform serving **36 REST endpoints across 8 workflow screens**; output feeds PwC audit work papers.
- Designed an **8-table normalized MySQL schema** with a polymorphic review-status table isolating UI writes from a quarterly ETL ingesting **19M+ raw general-ledger rows into 300K aggregated rows**, reconciled against Uber's public 10-Q within $2–3M tolerance.
- Eliminated a latent column-aliasing bug and raised new-line test coverage **34.6% → 100%** by migrating raw-SQL access to SQLAlchemy 2.0 ORM, shipped as a 3-PR stacked refactor.
- **Tech Used**: Python, FastAPI, SQLAlchemy, Pydantic, MySQL, Bazel, pytest, ruff, Redis, React (integration).

Menu Ingestion & Automation Platform — Uber Eats

- Cut vendor onboarding time **90% (24 h → 2 h)** and saved **$600K+ annually** by building an event-driven ingestion pipeline — Kafka event bus, **Apache Flink** real-time normalization, **Apache Spark** batch reprocessing — ingesting **30,000+ vendor menus/month** into Uber Eats.
- Raised extraction fidelity to **98% with 100% schema consistency** via an agentic AI pipeline (RAG + Gemini 2.5 Pro, SFT for schema adherence) for unstructured PDF/image menus.
- Increased successful ingestions by **95%** via anti-bot countermeasures (IP rotation, dynamic proxy pools); cut failure time-to-detect from hours to minutes with **Apache Pinot** real-time ingestion analytics and alerting.
- **Tech Used**: Python, Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, Selenium, Gemini, RAG, GCP, Docker.

Software Development Engineer 2 | MASTERS INDIA, Noida | December 2022 – June 2024

GST Compliance & E-Invoicing SaaS Platform

- Cut p95 API latency **75% (1.2 s → 300 ms)** and scaled to **1M+ daily transactions for 2,500+ enterprise clients** by leading the migration from a monolithic PHP (Laravel) app to asynchronous Python (FastAPI) microservices.
- Scaled bulk e-invoice processing to **100,000+ transactions per import** via chunked async workers and set-based database writes; implemented **encryption, RBAC, JWT auth, and audit logging** for financial-compliance data.
- Accelerated incident triage **70%** with centralized ELK logging and New Relic APM; raised **test coverage 35% → 82%** with 98% deployment success.
- **Tech Used**: Python, FastAPI, PostgreSQL, Redis, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer | GEEKSFORGEEKS, Noida | August 2021 – November 2022

Community & Courses Platform Backend

- Increased premium subscription sales **15–20%** by designing data models and REST APIs for engagement features while migrating the backend from PHP to Django (**100,000+ daily queries**); boosted course sales **30%** with a real-time influencer analytics dashboard.
- **Tech Used**: Python, Django, MySQL, Redis, REST, Cron, AWS.

Education & Achievements

B.Tech (Information Technology) | **Institute of Engineering & Technology, Lucknow** | **July 2017 – June 2021**